COUNTING TECHNIQUES FOR RATIONAL POINTS

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Lecture 1: The Hardy–Littlewood Circle Method

Motivation

Our first motivation comes from counting rational points of bounded height on a hypersurface $X \subseteq \mathbb{P}^n$. Say, for example

$$X = \{F(x_0, \dots, x_n) = 0\},$$

for some $F \in \mathbb{Z}[x_0, \dots, x_n]$ homogeneous. For $\mathbf{x} \in \mathbb{Z}^{n+1}$ with $\gcd(x_0, \dots, x_n) = 1$, consider the height function

$$H(\mathbf{x}) = \max_{0 \le i \le n} \{|x_i|\}.$$

Then

$$\#\{\mathbf{x} \in X(\mathbb{Q}) \ : \ H(\mathbf{x}) \le P\} = \frac{1}{2}\#\left\{(x_0, \dots, x_n) \in \mathbb{Z}^{n+1} \ : \ \begin{smallmatrix} \gcd(x_0,\dots,x_n)=1 \\ |x_i| \le P \ \forall i \\ F(x_0,\dots,x_n)=0 \end{smallmatrix}\right\},$$

where the $\frac{1}{2}$ comes from the fact that we can scale a points coordinates by -1 and have the same point.

Goal: We would like an asymptotic formula for this quantity as $P \to \infty$.

As a model example we consider the diagonal hypersurface

$$x_1^k + \cdots + x_s^k = x_{s+1}^k + \cdots + x_{2s}^k.$$

We will drop the gcd condition, since typically once we can count the set well without it then we can apply‘ mobius inversion’ to recover the size with it. Don't worry too much about this, we can discuss it in the exercise sessions.

Continuing with our model example, we take the notation:

Definition 1.

$$N(P) = \#\left\{(x_1, \dots, x_{2s}) \in \mathbb{Z}^{2s} \ : \ \begin{smallmatrix} |x_i| \le P \ \forall i \\ x_1^k + \cdots + x_s^k = x_{s+1}^k + \cdots + x_{2s}^k \end{smallmatrix}\right\}.$$

We want to understand $N(P)$ as $P \to \infty$.

Notation 2. For $\alpha \in \mathbb{R}$ set $e(\alpha) := e^{2\pi i \alpha}$. We define

$$T(\alpha) := \sum_{|x| \le P} e(\alpha x).$$

Note that for $n \in \mathbb{Z}$ we have

$$\int_0^1 e(\alpha n)\ d\alpha = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

Compute

$$\begin{aligned} \int_0^1 |T(\alpha)|^{2s}\ d\alpha &= \int_0^1 T(\alpha)^s T(-\alpha)^s d\alpha \\ &= \sum_{|x_1| \leq P} \sum_{|x_2| \leq P} \cdots \sum_{|x_{2s}| \leq P} \int_0^1 e(\alpha(x_1^k + \cdots + x_s^k - x_{s+1}^k - \cdots - x_{2s}^k) \\ &= N(P) \end{aligned}$$

This is the beginning of the circle method.

Heuristics for the growth of $N(P)$

Consider the map

$$\begin{aligned} \varphi : \left\{(x_1, \ldots, x_{2s}) \in \mathbb{Z}^{2s} \ : \ |x_i| \leq P\ \forall i\right\} &\to \left\{-2sP^k, \ldots, 0, 1, 2, \ldots, 2sP^k\right\} \\ (x_1, \ldots, x_{2s}) &\mapsto x_1^k + \cdots + x_s - x_{s+1}^k - \cdots - x_{2s}. \end{aligned}$$

If the images were roughly equidistributed, then we would expect something like

$$N(P) \sim P^{2s} P^{-k} = P^{2s-k}.$$

A second heuristic, $N(P) = \int_0^1 |T(\alpha)|^{2s}\ d\alpha$. For $\alpha = 0$, $P \in \mathbb{N}$, we have $T(0) = 2P + 1$. If $|\alpha| \leq c_0 P^{-k}$ for some sufficiently small c_0 then for $|x| \leq P$

$$\left|e(\alpha x^k) - 1\right| \leq \frac{1}{100},$$

and so we obtain $|T(\alpha)| \gg P$, where we are using the notation $a(P) \gg b(P)$ to mean that there exists a uniform constant c such that $a(P) \geq cb(P)$. From the region

$$\int_{|\alpha| \leq c_0 P^{-k}} |T(\alpha)|^{2s}\, d\alpha$$

we expect a contribution of size around P^{2s-k}.

For more 'generic' α we could expect $e(\alpha x^k)$ to be randomly distributed on the unit circle $\mathbb{S}^1 = \{u \in \mathbb{C}^\times \ : \ |u| = 1\}$. This would lead to the expectation $T(\alpha) \sim P^{1/2}$. Hence if we integrate over 'generic' α then we'd expect P^s contribution.

Hence, comparing these expectations, if $2s - k > s$ (i.e. $s > k$) then we expect the generic part to contribute to error and the expected main term should be of order P^{2s-k}.

Weyl's inequality and Hua's inequality

We are looking at

$$N(P) = \int_0^1 |T(\alpha)|^{2s}\, d\alpha,$$

where $T(\alpha) = \sum_{|x| \leq P} e(\alpha x^k)$.

Lemma 3 (Weyl's inequality)**.** *Let $\varepsilon > 0$, and assume that there are integers $a \in \mathbb{Z}$ and $q \in \mathbb{N}$ with $\gcd(a, q) = 1$ and*

$$\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{q^2}.$$

Then for $\kappa = 2^{k-1}$

$$|T(\alpha)| \ll P^{1+\varepsilon}\left(P^{-1/\kappa} + q^{-1/\kappa} + \left(\frac{P^k}{q}\right)^{-1/\kappa}\right).$$

Remark 4. The trivial bound is $|T(\alpha)| \ll P$. For $P \leq q \leq P^{k-1}$ we obtain

$$|T(\alpha)| \ll P^{1-\frac{1}{2^{k-1}}+\varepsilon}$$

Example 5. We now compare to quadratic Gauss sums. Let q be a prime number, $k = 2$, and $\alpha = \frac{1}{q}$ and $P = q$. Then

$$T\left(\frac{1}{q}\right) = \sum_{x=-q}^{q} e(x^2/q) = 1 + 2\sum_{1}^{q} e(x^2/q).$$

In particular, by bounding the Gauss sum, $T(1/q) \leq 2\sqrt{q} + 1$. I could instead have applied Lemma 3, which shows that

$$\left|T\left(\frac{1}{q}\right)\right| \ll q^{1+\varepsilon-1/2} \ll q^{1/2+\varepsilon},$$

so we are only off of the Gauss sum bound by an ε, which is not bad at all!

Now let's get an indication of how the proof of Weyl's inequality works. We can bound linear exponential sums

$$\sum_{|x|\leq P} e(\alpha x) \ll \min\left\{P, ||\alpha||^{-1}\right\},$$

where $||\alpha|| = \min_{m\in\mathbb{Z}} |\alpha - m|$.

Example 6 (Weyl differencing for $k = 2$)**.**

$$\begin{aligned}|T(\alpha)|^2 &= \left|\sum_{|x|\leq P} e(\alpha x^2)\right|^2 \\ &= \sum_{|x|\leq P}\sum_{|y|\leq P} e(\alpha(x^2 - y^2)).\end{aligned}$$

Substituting $x = y + h$ we obtain

$$\begin{aligned}|T(\alpha)|^2 &= \sum_{|y|\leq P}\sum_{\substack{h\in\mathbb{Z}\\ |y+h|\leq P}} e(\alpha(2hy + y^2)) \\ &\leq \sum_{h}\left|\sum_{\substack{|y|\leq P\\ |y+h|\leq P}} e(\alpha 2hy)\right|\end{aligned}$$

which is much easier and has reduced our initial degree to 1. Hopefully from this you can see how to Weyl difference your way down for arbitrary degree.